

TASK 3 - CUSTOMER CHURN PREDICTOR

Project Objective: Identify customers showing early signs of churn and rank them by risk using behavioural warning signals.

Methodology

The Customer Churn Predictor was developed using Python and pandas in Google Colab to identify customers who may be at risk of leaving based on changes in recent customer behaviour.

The first churn signal identifies customers who have been inactive for 30 days or more. Days inactive were calculated by comparing each customer's last activity date with the analysis date. Customers with 30 or more inactive days were flagged.

The second churn signal identifies customers whose order frequency decreased by 50% or more. Recent 30-day orders were compared with the previous 30-day period, and the percentage decline was calculated for each customer.

A simple risk score was created by assigning one point for each triggered signal. A score of 0 represents Low Risk, a score of 1 represents Medium Risk, and a score of 2 represents High Risk.

Customers with a risk score above zero were included in the final at-risk list. They were ranked first by Risk Score, then by Days Inactive, and finally by Order Drop Percentage, all from highest to lowest. This places the strongest churn warning signs at the top of the list.

Ranked List of At-Risk Customers

Ranking priority: Risk Score -> Days Inactive -> Order Drop %.

Total At-Risk 23	High Risk 5	Medium Risk 18
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Rank	Customer ID	Customer Name	City	Days Inactive	Order Drop %	Risk Score	Risk Level
1	C028	Victoria Ansah	Accra	80	80.00%	2	High Risk
2	C007	Akosua Nyarko	Koforidua	80	50.00%	2	High Risk
3	C025	Isaac Danso	Accra	39	50.00%	2	High Risk
4	C020	Richard Boadu	Accra	36	87.50%	2	High Risk
5	C040	Mercy Koomson	Accra	36	71.43%	2	High Risk
6	C021	Angela Darko	Tema	76	40.00%	1	Medium Risk
7	C035	David Mensima	Accra	66	25.00%	1	Medium Risk
8	C014	Samuel Ofori	Accra	47	20.00%	1	Medium Risk
9	C010	Nana Frimpong	Kumasi	44	33.33%	1	Medium Risk
10	C015	Grace Amankwah	Kumasi	40	11.11%	1	Medium Risk
11	C030	Mary Osei	Accra	33	11.11%	1	Medium Risk
12	C005	Efua Addo	Accra	31	40.00%	1	Medium Risk
13	C012	Daniel Tetteh	Tamale	25	81.82%	1	Medium Risk
14	C018	Emmanuel Kwarteng	Sunyani	23	81.82%	1	Medium Risk
15	C027	Stephen Addai	Tema	23	75.00%	1	Medium Risk
16	C009	Adwoa Asante	Tema	17	75.00%	1	Medium Risk
17	C016	Michael Quaye	Tema	16	60.00%	1	Medium Risk
18	C029	Francis Adjei	Kumasi	14	50.00%	1	Medium Risk
19	C024	Patricia Baah	Takoradi	9	85.71%	1	Medium Risk
20	C032	Ruth Boateng	Accra	8	90.00%	1	Medium Risk
21	C036	Beatrice Tandoh	Kumasi	8	75.00%	1	Medium Risk
22	C008	Kofi Osei	Accra	5	66.67%	1	Medium Risk
23	C004	Yaw Boateng	Takoradi	3	60.00%	1	Medium Risk

Conclusion: The predictor successfully flags and ranks customers with measurable churn warning signs, allowing retention efforts to focus on the highest-risk customers first.